Doubt Session

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Q: add two binary numbers & give the result in decimal.

→ 1) write the function to convert binary to decimal.
↳ without using pow fn.

```
public class Main {

    public static void main() {

    }
```

```
Public static int binToDecimal(int bin){
    int num = 0;
    int base = 1 ....> 2^0
    while (bin > 0) {
        int multiplier = bin % 10;
        num = num + multiplier x base;
        base = base x 2; //To increment base from the power of 2
        bin = bin / 10;
    }
    return num;

Public static void main() {
    int bin1 = 100110 , bin2 = 10001;
    int dec1 = binToDecimal(bin1);
    int dec2 = binToDecimal(bin2);
    Sout(dec1+dec2);
}
```

Q: Write a function to calculate factorial of a number (return an int).

code

```
public static int fact(int n){
    int num = 1;
    for (int i=1 ; i<=n ; i++){
        num = num * i;
    }
    return num;
}
```

i=1 ; i<=n

Q. Pascal's triangle.

```
0 1 2
1                 → row = 0
1 1               → row = 1
1 2 1             → 2
1 3 3 1           → 3
1 4 6 4           → 4
```

row = 4
Col = 2

$^{4}C_{2} = 6$

$\frac{4!}{2! \times 2!}$

$= \frac{4 \times 3 \times 2!}{2! \times 2!}$

$= \frac{4 \times 3}{2}$

$= 6$

row = 3
col = 1

$^{row-1}C_{col-1}$

$^{0}C_{0} = 1$

$^{n}C_{0} = 1$

$^{row}C_{col}$

formula

$i=1 ; i<=0$

$\boxed{{}^{row}C_{col}}$

$\boxed{{}^{1}C_0 = 1}$

col = 1

$\boxed{{}^{3}C_1 = 3}$

$${}^{n}C_r = \frac{n!}{r! \times (n-r)!}$$

$${}^{5}C_2 = \frac{5!}{(5-2)!\,2!}$$

$$= \frac{5!}{3! \times 2!}$$

$$= \frac{5 \times 4 \times \cancel{3!}}{\cancel{3!} \times 2!} = \frac{5 \times 4}{2 \times 1} = 10$$

$\boxed{n=3}$

0/p 1 3 3 1

0 1 2 3

${}^{3}C_0 = 1$

```
for (int j=0; j<=n; j++){
    Sout( fact(n)/(fact(n-j)*fact(j)));
}
```

fact(0); → 1

fact(3)/fac(3) × fact(0) = 1

```
    1
   1 1
  1 2 1
 1 3 3 1
1 4 6 4 1
```

n=0 , n=1 , n=2

$n \rightarrow \boxed{0 \rightarrow 4}$

$\boxed{\frac{fact(n)}{fact(n-j) \times fact(j)}}$ 0

n = 3 , 4 , 13

n

n → not constant

n=0
n=1
n=2
n=4 → total rows

?

n = 1
n = 2

```
int m = 4;  → total rows
for (int n = 0; n <= m; n++){
    for (int j = 0; j <= n; j++){
        sout (fact(n)/(fact(j) x fact(n-j)));
    }
    Sout ("\n");
}
```

Q. Decimal to Binary Conversion:

→ Code!!

→ 43 → ....

2	43	1
2	21	1
2	10	0
2	5	1
2	2	0
	1	

2) 43 (21
42
1

2) 21 (10
20
1

101011 = 43

→ 68

2	68	0
2	34	0
2	17	1
2	8	0
2	4	0
2	2	0
2	1	

→ 00 10001

2) 68 (34
68
0

2) 34 (17
34
0

1000100

1000100 = 68

$2^6 + 2^2 = 64 + 4$
$= 68$

8 = 1000

```
int ans=0;
While (num > 0){
    num/.2;
```

10

```
int ans = 0;
while (num > 0){
    int rem = num % 2;
    ans = ans * 10 + rem;

    num = num/2;
}
int rev = 0;    while (
```

```
int n = sc.nextInt();
 int ans = 0;
while (n>0){
 int rem = n%2;
ans = ans * 10 + rem;
n = n/2;
}
 System.out.println(ans);
int rev = 0;
while(ans > 0)
{
 int digit = ans % 2;
rev = rev * 10 + digit;
 ans = ans/10;
 }
System.out.println(rev);
} }
```